DGMS 2019 • SECTION E • QUESTION 30

Sedimentary Dynamics: Bedform Succession with Increasing Bed Shear Stress & Flow Regimes

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Physical Sedimentology & Hydrodynamics Guide

Question Statement

Q.30. Which one of the following sequences of sedimentary structures is correct in terms of increasing bed shear?

a (a) Parting lineation, dune, flute mark, ripple

b (b) Ripple, dune, parting lineation, flute mark

c (c) Dune, parting lineation, flute mark, ripple

d (d) Flute mark, parting lineation, ripple, dune

Correct Answer: (b) Ripple, dune, parting lineation, flute mark

INCREASING BED SHEAR STRESS (τ_0):

Current Ripple → Subaqueous Dune → Parting Lineation (Upper Plane Bed) → Flute Mark (Turbulent Scour)

Official Textbook Figures (Sam Boggs Jr., Principles of Sedimentology & Stratigraphy)

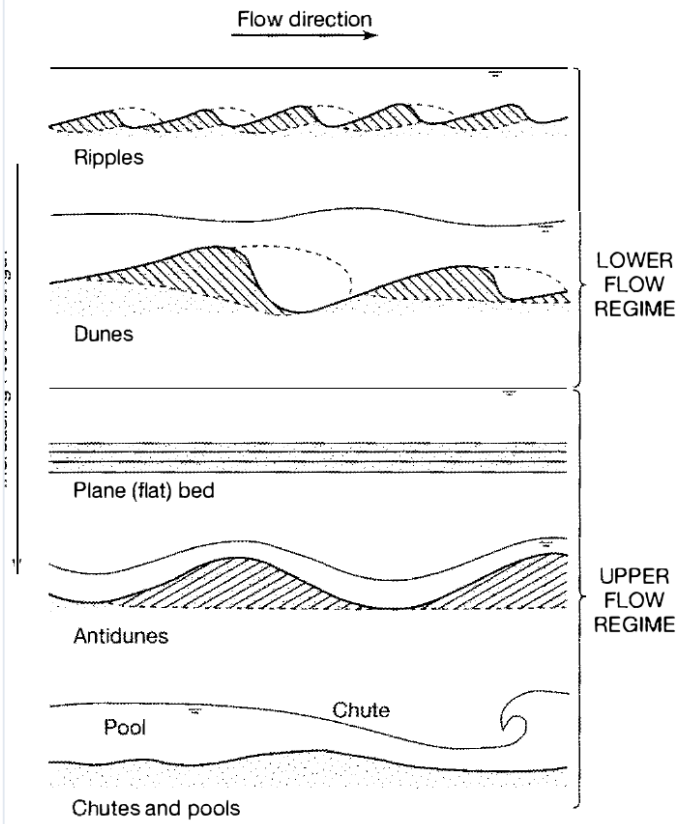


Figure 4.9
The succession of bedforms that develops during unidirectional flow of sandy sediment (0.25–0.7 mm) in shallow water as flow velocity increases. [Modified slightly from Blatt, H., G. V. Middleton, and R. Murray, 1980, Origin of sedimentary rocks, 2nd ed., Fig. 5.3, p. 137. Reprinted by permission of Prentice-Hall, Englewood Cliffs, N.J.]

anging in size from silt (0.06 mm) to sand as coarse as 0.7 mm. Larger bedforms ith spacing, or wave length, ranging from under 1 m to over 1000 m are called **unes** (Ashley, 1990). Dunes are similar in general appearance to ripples except r size. They form at higher flow velocities in sediment ranging in grain size from ne sand to gravel. The ripple index of dunes ranges from about 5 in finer sands to

Figure 1: Sam Boggs Jr., Chapter 4, Figure 4.9. The succession of bedforms that develops during unidirectional flow of sandy sediment (0.25–0.7 mm) as flow velocity and bed shear stress increase. From bottom to top: **Ripples → Dunes → Upper Plane Bed (Parting Lineation) → Antidunes / Chutes & Pools**.

e ridges and hollows is commonly on the order of the diameter of the sandstone ains. The grains in the sandstone generally have a mean orientation of their long es parallel to the lineation. The lineation is oriented parallel to current flow, and us its presence in ancient sandstones is useful in paleocurrent studies, although shows only that the current flowed parallel to the parting lineations and does ot show which of the two diametrically opposed directions was the flow direc- on. Parting lineation occurs in newly deposited sands on beaches and in fluvial vironments. It is most common in ancient deposits in thin, evenly bedded sand- ones. Its origin is obviously related to current flow and grain orientation, proba- y owing to flow over upper-flow regime plane beds, but the exact mechanism by hich parting lineation forms is poorly understood.



Figure 2: Sam Boggs Jr., Chapter 4, Figure 4.54. Parting lineation (primary current lineation) formed on bedding planes of flat-laminated sandstones deposited under the **upper flow regime** at high bed shear stress.

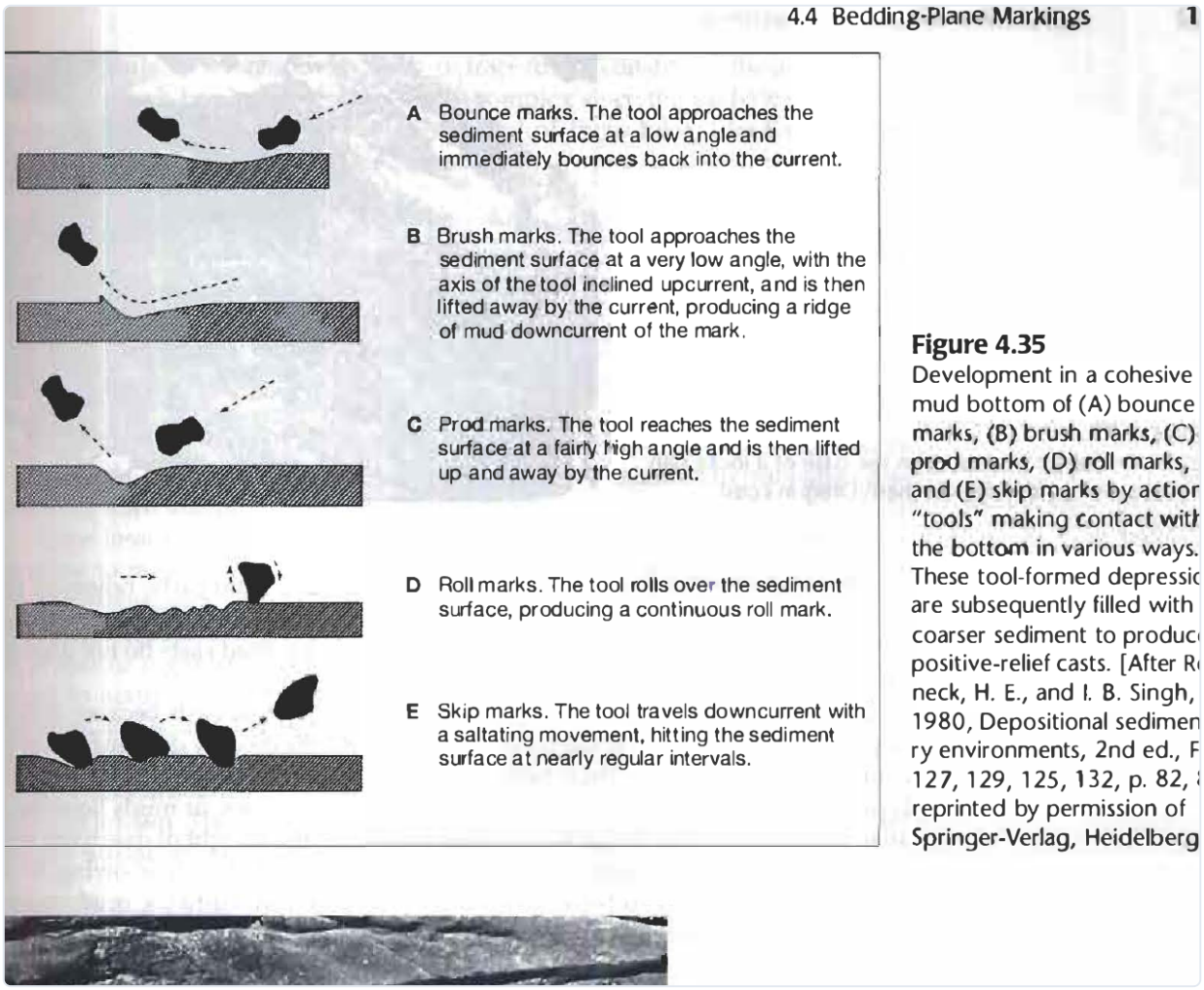


Figure 3: Sam Boggs Jr., Chapter 4, Figure 4.36. Flute casts on the sole of a turbidite sandstone bed. Flutes represent the **highest bed shear and turbulent scour** produced by concentrated horizontal fluid vortices eroding into cohesive mud before sand deposition.

In-Depth Option Evaluation & Hydrodynamic Progression

(b) Ripple, dune, parting lineation, flute mark — [CORRECT]

Hydrodynamic Succession with Increasing Bed Shear ($\tau_o = \rho C_d U^2$):

- **1. Ripples (Lowest Bed Shear):** Form just above the threshold velocity for sediment motion in the Lower Flow Regime. Small-scale bedforms ($\lambda < 20$ cm, $H < 4$ cm) where grains saltate over the stoss side and avalanche down the lee slope.
- **2. Dunes (Moderate Bed Shear):** Form at higher velocity and stream power in the Lower Flow Regime. Large-scale bedforms ($\lambda > 0.6$–1 m, $H > 5$ cm) with active flow separation and wake vortices over the crest.

- **3. Parting Lineation / Upper Stage Plane Bed (High Bed Shear):** Upper Flow Regime (Froude number Fr ≈ 1). Dunes are washed out and sheared flat into flat laminae. High-speed and low-speed streaks generated by streamwise turbulent boundary-layer vortices align elongate grains into parting step lineations.
- **4. Flute Marks (Highest Bed Shear & Extreme Turbulent Scour):** Produced by hyper-turbulent erosive vortices at the head of high-energy flows (e.g., turbidity currents or catastrophic surges) scouring directly into the bed before deposition can begin. Basal shear stress exceeds even that of plane-bed deposition!

(a, c, d) Incorrect Permutations — [INCORRECT]

• **(a)** Starts with Parting lineation (Upper flow regime) followed by Dune (Lower flow regime) — non-monotonic shear order.
• **(c)** Places Ripple (lowest shear) at the very end after Flute mark (highest shear) — inverted.
• **(d)** Inverted sequence (from highest to lowest shear: Flute mark ➔ Parting lineation ➔ Ripple/Dune), but the question explicitly demands **increasing bed shear**!

Comprehensive Hydrodynamic & Bedform Parameters

Bedform / Structure	Flow Regime	Froude Number (Fr)	Relative Bed Shear (τ_o)	Internal Stratification Produced
Current Ripples	Lower Flow Regime	Fr << 1 (0.1–0.3)	Lowest (Just above critical threshold τ_{cr})	Small-scale planar and trough cross-lamination (ripple cross-laminae)
Subaqueous Dunes	Lower Flow Regime	Fr < 1 (0.3–0.8)	Moderate	Medium- to large-scale tabular and trough cross-bedding
Upper Plane Bed (Parting Lineation)	Upper Flow Regime	Fr ≈ 0.8–1.2	High (Dunes planed off; intense sheet flow)	Parallel horizontal plane lamination, parting current lineation
Flute Marks (Turbulent Scour)	**Erosional High-Energy Regime**	**Fr > 1 (Highly turbulent flow head)**	**Highest (Violent boundary-layer vortex cavitation and erosion)**	**Sole marks on base of thick graded turbidite beds (T_a division)**

Direct Textbook Quotations & Citations

Sam Boggs Jr. (2006), *Principles of Sedimentology and Stratigraphy, 4th ed., Chapter 4, pp. 83–85:*

"Numerous flume experiments have established that under unidirectional flow in sand, **the succession of bedforms that develops as flow velocity and bed shear stress increase is: ripples → dunes → plane bed (with parting lineation) → antidunes**.*"*

— Sam Boggs Jr., Chapter 4, p. 83

J. D. Collinson & D. B. Thompson (1989), *Sedimentary Structures, 2nd ed., Chapter 4 & 5:*

"Flutes are produced by localized turbulent scour of a cohesive mud bed by fluid vortices generated in high-velocity, high-shear flows prior to deposition... In terms of boundary shear stress, flute erosion represents the upper boundary of flow power, exceeding the threshold required for plane-bed transport."

— Collinson & Thompson (1989)

Exam Traps & Rapid Elimination Strategy

Universal Flow Regime Rule:

1. Always anchor the two endpoints:
   - **Lowest Shear: Ripples** (appear immediately when sand starts moving).
   - **Highest Shear / Scour: Flute Marks** (requires extreme turbulence to scour cohesive mud before sand settles).
2. Between Ripples and Flutes: **Dunes** (Lower Flow Regime) come before **Parting Lineation** (Upper Flow Regime plane bed).
3. Therefore, order: **Ripple → Dune → Parting Lineation → Flute Mark**!